WORKED SOLUTION

STEP III 1999, Question 7

Pure • Groups and Isomorphisms

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q7

The key idea

The expression $x + y + axy$ is very nearly a product. Multiplying by a and adding 1 gives

$$1 + a(x * y) = 1 + ax + ay + a^2xy = (1 + ax)(1 + ay),$$

so under the change of variable $X = 1 + ax$ the operation $*$ is just ordinary multiplication in disguise. Every group property then comes from the corresponding property of multiplication, and the one number we must throw out is the one that corresponds to $X = 0$.

Solution

Throughout, write $\phi(x) = 1 + ax$. Since $a \neq 0$, ϕ is a bijection from $\mathbb{R}$ to $\mathbb{R}$, so two real numbers u, v are equal exactly when $\phi(u) = \phi(v)$. The identity above reads

$$\phi(x * y) = \phi(x)\,\phi(y) \qquad \text{for all real } x, y. \tag{†}$$

Associativity

Using (†) twice on each side,

$$\phi((x * y) * z) = \phi(x * y)\,\phi(z) = \phi(x)\,\phi(y)\,\phi(z),$$
$$\phi(x * (y * z)) = \phi(x)\,\phi(y * z) = \phi(x)\,\phi(y)\,\phi(z).$$

These are equal because multiplication of real numbers is associative, and since ϕ is one-to-one, $(x * y) * z = x * (y * z)$. So $*$ is associative.

(If you prefer a direct check: both $(x * y) * z$ and $x * (y * z)$ expand to the symmetric expression $x + y + z + a(xy + yz + zx) + a^2xyz$.)

The group

Ask yourself Which number has to be removed, and why can't we keep all of $\mathbb{R}$?

Under ϕ, the number $x = -\frac{1}{a}$ corresponds to $\phi(x) = 0$, and 0 has no multiplicative inverse. Concretely, for any real y,

$$\left(-\tfrac{1}{a}\right) * y = -\tfrac{1}{a} + y - y = -\tfrac{1}{a},$$

so $-\frac{1}{a}$ absorbs everything and can never be combined with anything to give the identity. So we take

$$G = \mathbb{R} \setminus \left\{-\tfrac{1}{a}\right\},$$

which is exactly the set of x with $\phi(x) \neq 0$. We check the four axioms.

Closure. If $x, y \in G$ then $\phi(x) \neq 0$ and $\phi(y) \neq 0$, so by $(\dagger)$, $\phi(x * y) = \phi(x)\phi(y) \neq 0$. Hence $x * y \neq -\frac{1}{a}$, and $x * y$ is certainly real, so $x * y \in G$.

Associativity. This was shown above for all real numbers, so in particular for elements of G.

Identity. We have $x * 0 = x + 0 + 0 = x$ and $0 * x = x$ for every x, and $0 \in G$ since $\phi(0) = 1 \neq 0$. So 0 is the identity. (This matches $\phi(0) = 1$, the multiplicative identity.)

Inverses. Given $x \in G$, we want y with $x * y = 0$, that is $x + y(1 + ax) = 0$. Since $1 + ax \neq 0$ we may divide, giving

$$x^{-1} = \frac{-x}{1 + ax}.$$

This lies in G, because

$$\phi\left(\frac{-x}{1+ax}\right) = 1 - \frac{ax}{1+ax} = \frac{1}{1+ax} \neq 0,$$

and since $*$ is commutative ($x * y = y * x$ is clear from the formula), $x^{-1} * x = 0$ as well.

All four axioms hold, so $(G, *)$ is a group (indeed an abelian one), where the excluded number is $-\frac{1}{a}$.

A subgroup of order 2

The multiplicative group of non-zero reals has the subgroup $\{1, -1\}$. Pulling this back through ϕ: $\phi(x) = 1$ gives $x = 0$, and $\phi(x) = -1$ gives $x = -\frac{2}{a}$. So we try

$$H = \left\{0, -\tfrac{2}{a}\right\}.$$

These are two different elements of G (as $a \neq 0$, $-\frac{2}{a} \neq 0$ and $-\frac{2}{a} \neq -\frac{1}{a}$). Directly,

$$\left(-\tfrac{2}{a}\right) * \left(-\tfrac{2}{a}\right) = -\tfrac{4}{a} + a \cdot \tfrac{4}{a^2} = 0,$$

and $0 * 0 = 0$, $0 * \left(-\frac{2}{a}\right) = -\frac{2}{a}$. So H is closed, contains the identity, and each element is its own inverse. Hence H is a subgroup of $(G, *)$ with exactly two elements.

In fact it is the only one: a group of order 2 is $\{e, h\}$ with $h * h = e$ and $h \neq e$, and $h * h = 0$ means $h(2 + ah) = 0$, so $h = -\frac{2}{a}$.

Checks

Take $a = 1$. Then $x * y = x + y + xy = (1 + x)(1 + y) - 1$, the excluded number is -1, and the order-2 subgroup is $\{0, -2\}$: indeed $(-2) * (-2) = -4 + 4 = 0$. Also $2 * 3 = 11$ and $(1 + 2)(1 + 3) = 12 = 1 + 11$, as $(\dagger)$ predicts. With a computer we verified, in exact rational arithmetic on about 5000 random choices of a, x, y, z, the identity $(\dagger)$, associativity, closure in G, the identity 0, the inverse formula, and closure of $\{0, -\frac{2}{a}\}$.

Common slips

- Forgetting to check that the inverse $\frac{-x}{1+ax}$ is itself in G, or that $x * y$ never equals the excluded number. Closure is the axiom that justifies removing exactly one point.
- Offering $\{0\}$ or $\{0, -\frac{1}{a}\}$ as the subgroup: the first has one element, and $-\frac{1}{a}$ is not in G at all.

Beyond STEP

The map $\phi(x) = 1 + ax$ is an *isomorphism* from $(G, *)$ to the group $(\mathbb{R} \setminus \{0\}, \times)$: a bijection that turns one operation into the other. Once you spot an isomorphism, every structural fact transfers for free, which is why $(G, *)$ has exactly one element of order 2, just as $\times$ has only -1. Isomorphisms, and the idea that two groups can be "the same" while looking different, are central to a first course in abstract algebra; Allenby's *Rings, Fields and Groups* is a good place to read about them.

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